

GPUs and Model Acceleration

Choose your GPU: plan your team's training run and deployment. Show every calculation. Work in pairs!

GROUP MEMBER 1 _____

GROUP MEMBER 2 _____

Your team: A startup in Cape Town building an LLM for a customer service assistant that classifies support tickets by urgency and responds to customers.

The job: Train a 1B parameter LLM on 20B tokens in bfloat16 with the Adam optimizer, then deploy it.

1B

PARAMETERS

20B

TRAINING TOKENS

12 GBTRAINING
ACTIVATIONS**\$1,000**

BUDGET

72 hDEADLINE (3
DAYS)**GPU MENU**

You can rent more than one of any GPU.

GPU	VRAM	SPEED	PRICE PER HOUR
A	16 GB	100 TFLOPS	\$1
B	40 GB	300 TFLOPS	\$4
C	80 GB	1,000 TFLOPS	\$10

HINT Assume FLOPs = $6 \times N \times D$

N = parameters, D = training tokens. Assume 100% utilization and perfect scaling across GPUs.

 $1B = 10^9 \cdot 1 \text{ TFLOPS} = 10^{12} \text{ FLOPS} \cdot 1 \text{ hour} = 3,600 \text{ s}$ **1 Which GPUs can train your model?**

Calculate the memory needed for training. Which GPUs can fit it as is?

PARAMETERS		GRADIENTS		OPTIMIZER		ACTIVATIONS		TOTAL
	GB	+		GB	+		GB	= GB

GPU	VRAM	FITS AS IS?	NEEDS AN OPTIMIZATION?
A	16 GB	☐ Yes ☐ No	☐ Yes ☐ No
B	40 GB	☐ Yes ☐ No	☐ Yes ☐ No
C	80 GB	☐ Yes ☐ No	☐ Yes ☐ No

Workings and reflection on the next page →

1 Which GPUs can train your model? (continued)

WORKINGS



Reflect

Which part of training memory would shrink if you halved the batch size, and which parts would not?

2 What would each option cost, and how long would it take?

Calculate the training time and cost on one of each GPU. Which options meet both limits?

TOTAL COMPUTE _____ FLOPs

GPU	TIME (HOURS)	COST (\$)	MEETS BOTH LIMITS?
A			☐ Yes ☐ No
B			☐ Yes ☐ No
C			☐ Yes ☐ No

WORKINGS



Reflect

Why isn't the cheapest GPU per hour necessarily the cheapest training option?
What matters more for total cost: GPU price or training time? Why?

3 What is your final training plan?

Choose your GPU, how many, and which optimizations. You may pick a cheaper GPU if you make it work with the techniques from the lecture.

GPU	HOW MANY	OPTIMIZATIONS	GB PER GPU	HOURS	COST
					\$

MEETS THE DEADLINE? ☐ Yes ☐ No

MEETS THE BUDGET? ☐ Yes ☐ No

WORKINGS



Reflect

What trade-off did you make to meet both the deadline and budget? Why do you think the trade-off makes sense?

4 How much memory does inference need?

Your model is trained. Calculate the memory needed to run it in bfloat16 (bf16) and in int4, then compare with training.

At inference: no gradients or optimizer states. **Temporary activations:** 1 GB. **int4:** 0.5 bytes per parameter. The KV cache and all other memory are ignored.

SETTING	PARAMETERS	TEMPORARY ACTIVATIONS	TOTAL	FITS ON GPU A?	TIMES LESS THAN TRAINING
Training				☐ Yes ☐ No	1×
Inference: bf16		1 GB		☐ Yes ☐ No	
Inference: int4		1 GB		☐ Yes ☐ No	

Which GPU and format would you use to serve customers? Why?



Reflect

What might you sacrifice by choosing the lowest-precision option (int4) for inference?